A Reflexive Non-B-Convex Subspace of a Variable Lebesgue Space

Math discovery packet

June 16, 2026

1 Source Question

The source paper is

> C. S. Barroso, ℓ_1 *spreading models and the FPP for Cesaro mean nonexpansive maps*, arXiv:2403.18113.

On page 12, in Section 5(h), the source defines variable Lebesgue spaces with measurable exponent

$$p : \Omega \to [1, +\infty]$$

and asks whether there is a reflexive non-B-convex subspace of some $L^{p(\cdot)}(\Omega)$.

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Corollary 5.4. *Let (Ω, Σ, μ) be an arbitrary σ-finite measure space and let $p\colon \Omega \to [1, +\infty]$ be a measurable function. If $L^{p(\cdot)}(\Omega)$ is reflexive then it has the fixed point property for $\mathfrak{cm}$-nonexpansive maps.*

We note that this result yields a slight generalization of [17, Theorem 4.2], as no further condition is needed on the function $p(\cdot)$. As highlighted in [17, Theorem 3.3], see also comments in p. 12, one can find plenty of examples of nonreflexive VLSs that still have the weak-FPP. For instance, $L^{1+x}([0,1])$ is one of them. All these remarks naturally lead us to ask whether there is a reflexive *non* B-convex subspace of $L^{p(\cdot)}(\Omega)$.

2 Result

Theorem 1. *As the question is written, the answer is affirmative. There is a complete σ-finite measure space (Ω, Σ, μ), a measurable exponent $p : \Omega \to [1, +\infty]$, and a closed linear subspace*

$$Y \subset L^{p(\cdot)}(\Omega)$$

such that Y is reflexive and non-B-convex.

3 Idea of the Proof

The endpoint $p = \infty$ is allowed in the source's definition. With $p(\cdot) \equiv \infty$, the variable Lebesgue space is simply an L_∞ space. But L_∞ spaces are universal for separable Banach spaces: every separable Banach space embeds linearly isometrically into $L_\infty(K, \mu)$ for a compact metric K and a full-support probability measure μ. The source already points to reflexive non-B-convex examples, namely the Cesaro sequence spaces ces_p, $1 < p < \infty$.

4 Proof

The source recalls, citing Astashkin–Maligranda, that the Cesaro sequence spaces ces_p, $1 < p < \infty$, are reflexive and non-B-convex. Fix one such space, say

$$E = \mathrm{ces}_2 .$$

This is a separable Banach sequence space; for instance, truncations of a vector converge in the Cesaro norm by monotone convergence applied to the defining p-summable Cesaro averages.

Let $K = B_{E^*}$ with the weak-star topology. Since E is separable, K is compact metrizable. The canonical evaluation map

$$J : E \to C(K), \qquad (Jx)(x^*) = x^*(x),$$

is a linear isometry by the Hahn–Banach theorem.

Choose a probability measure μ on K with full support; for example, put positive summable masses on a countable dense subset of K. Then for every $f \in C(K)$,

$$\|f\|_{C(K)} = \|f\|_{L_\infty(K,\mu)}.$$

Indeed, the inequality $\|f\|_{L_\infty} \le \|f\|_\infty$ is immediate. For the reverse inequality, if $|f(t_0)| > \|f\|_\infty - \varepsilon$, continuity and full support give a positive-measure neighborhood on which $|f| > \|f\|_\infty - 2\varepsilon$.

Now take

$$\Omega = K, \qquad p(t) \equiv \infty.$$

The modular definition in the source gives

$$L^{p(\cdot)}(\Omega) = L_\infty(K, \mu)$$

with its usual essential-supremum norm. Therefore $J(E)$ is a closed linear subspace of $L^{p(\cdot)}(\Omega)$, isometric to $E = \mathrm{ces}_2$. Consequently $J(E)$ is reflexive and non-B-convex.

5 Limitations

This answers the literal formulation, because the source explicitly permits the value $+\infty$ in the exponent range. It does not settle a stricter variant in which the exponent must be finite-valued, or in which the ambient variable Lebesgue space is required to satisfy additional nondegeneracy conditions.

6 Novelty Check

The run's cheap indexes were searched for arXiv:2403.18113, variable Lebesgue, non-B-convex subspace, Cesaro mean nonexpansive maps, and nearby FPP/spreading-model terminology. No prior packet or attempt for this source question was found. Bounded web searches for exact phrases including "reflexive non B-convex subspace", "variable Lebesgue", and "$L^{p(\cdot)}$" did not return an already-recorded answer to this literal endpoint observation.

7 References

- C. S. Barroso, *ℓ_1 spreading models and the FPP for Cesaro mean nonexpansive maps*, arXiv:2403.18113.
- S. V. Astashkin and L. Maligranda, *Cesaro function spaces fail the fixed point property*, Proc. Amer. Math. Soc. 136 (2008), 4289–4294.

8 Human Review Recommendation

Review as a literal-formulation full answer. The main point to check is whether the source intended to exclude the endpoint $p = \infty$; if so, this packet should be reclassified as an endpoint observation rather than a full answer to the intended finite-exponent variant.